WEB READINESS FOR AI AGENTS

Vedang Ratan Vatsa
veda.ng

ABSTRACT

AI agents use AI models and software tools to carry out tasks, including reading websites and interacting with online services. This retrospective study examines recorded results from 61 automated checks across 50,000 domains sampled from three popularity groups in the Tranco ranking. The final records cover September 3-4, 2026. The request procedure excludes logging in and running page JavaScript. All percentages here describe the sample. The text rule requires more than 300 characters after removing scripts, styles, and HTML tags; this rule passed on 45.71% of domains. On 30.16% of all domains, the text rule passed and both requests using the crawler names GPTBot and PerplexityBot were recorded as HTTP 200 ("OK"). Checks for sitemap and robots-file markers and machine-readable page metadata (JSON-LD) passed on 27.37% and 21.37%, respectively. At least one of three checks for llms.txt guidance files, Markdown-formatted content, and basic Model Context Protocol (MCP) interface markers passed on 7.50% of domains. A check for an OpenAPI version marker in candidate API descriptions passed on 0.16%. The content and discovery checks recorded more matches than the tested interface checks. These findings describe the recorded public signals; they do not establish an overall readiness level or agent-task success rate.

Keywords: AI agents, agent readiness, web measurement, robots.txt, llms.txt, Model Context Protocol, OpenAPI, Tranco, CrUX

1 INTRODUCTION

AI agents are software systems in which an AI model chooses actions and uses tools to work toward a user's goal. Those tools can include a web browser, a search service, or an application programming interface (API), which lets software request data or operations from another service. An agent can use the result of one action to decide what to do next [31]. For example, it may search for product information, compare alternatives, or complete an online form.

In July 2025, OpenAI introduced ChatGPT agent with tools for browsing websites and using APIs [32]. Such products make website readiness a practical question for users and developers. A user can ask an agent to complete a task, but the agent still needs to find the right information and a way to use the relevant service.

Websites support these activities in different ways. A page may contain useful information without providing a documented API. An API may offer useful operations without a homepage containing much text. Some agents use a full browser, while others request pages and files directly. This study asks what content and signs of service interfaces can be found through direct web requests, without logging in or running page JavaScript.

This retrospective analysis asks how often the selected rules match, how their matches overlap, and how the results depend on rank and collection procedure. It combines complete check counts with rank-adjusted estimates, paired repeat-scan comparisons, and tests with prepared responses. The contribution is a reproducible description of these public-web signals and the limits on interpreting them as readiness; completed agent tasks were not part of the scan.

2 Related Work

WebArena tests whether agents complete tasks on a controlled set of websites. It checks whether their answers and website changes match what the task asked for [33]. This study counts matches to website checks across public domains. A percentage of domains passing a check is a different measurement from a percentage of tasks an agent completes.

Web surveys also count different things. HTTP Archive records how pages are built and which web technologies they use [21]. Thunderbit reports 586 llms.txt files passing its validation among 10,000 Tranco domains, compared with 1,606 domains returning HTTP 200 at the tested address [22]. Many replies led elsewhere or contained an ordinary web page instead of the requested file. Reaching an address was not enough to establish that the intended file was returned.

Other studies read robots.txt, a file containing a website's published rules for web crawlers [23; 24; 25]. This study also sends page requests using the names GPTBot and PerplexityBot and records whether the server returns HTTP 200 ("OK"). A published rule permitting crawling does not guarantee that a request succeeds. These are test requests carrying crawler names, not visits by the companies' own crawlers.

Publishing a file does not guarantee that anyone uses it. Ahrefs reports that 97% of detected llms.txt files among 137,210 domains using its Web Analytics service received no requests in May 2026 [26]. Ora's methodology page lists 126 checks across four groups as of October 2026 [29]. These studies examine different sets of websites and use different rules, so their percentages are not directly comparable. This study brings several kinds of checks together to examine content access, resource discovery, and possible service interfaces.

3 Study Design

Web readiness depends on the task and how an agent uses a site. This study examines recorded check labels related to three activities. Access checks concern response codes and text length. Discovery checks look for resource lists and descriptions. Service-interface checks look for API and protocol information, access-permission details, and payment-related text. A marker is a word or field sought by a rule. The measured outcome is a match to that rule, rather than verified support for the activity.

The result is a set of observations for each domain. There is no single pass mark that makes a website ready for every agent. An agent need not use every format or protocol, and optional files such as llms.txt are not requirements for all tasks.

3.1 Sample

The sample comes from the first million entries of Tranco list ZJQ6G, generated on 2 September 2026 [1; 27; 28]. The list combines five ranking providers over 30 days.

The sample includes every domain ranked 1 through 10,000. From each of the next two rank groups, it includes 20,000 domains picked at random, with no domain picked twice. The random seed is 20260903, allowing the same selection to be reproduced.

Table 1: Rank groups in the sampling frame and study sample.

Rank range	Domains in full list	Domains sampled
1-10,000	10,000	10,000
10,001-100,000	90,000	20,000
100,001-1,000,000	900,000	20,000

Each listed hostname is one observation. Several names can lead to the same website, so the counts do not represent unique organizations or website owners. The unadjusted percentages describe this sample; Section 4.3.1 gives estimates adjusted to the rank mix of the full list.

The study also checks whether each domain appears in the July 2026 Chrome User Experience Report (CrUX) top-million list of web addresses [2; 3]. For this comparison, it ignores the URL

protocol and port number and treats www.example.com and example.com as the same hostname. It finds 23,592 matches. CrUX groups addresses into broad popularity bands and applies rules about popularity and being findable on the public web. CrUX is also one of the sources used to build Tranco [28].

3.2 Why these 61 checks

The retained checklist examines public pages, files, and server responses. The reason for examining these signals is their connection to how agents use websites. Depending on the task, agents need to obtain information, find relevant resources, or use services [31]. Access tests, resource lists, and interface descriptions address those needs. General metadata and security headers provide context, while payment-related signals matter only for some tasks.

The checklist uses public URLs and server replies without requiring accounts, browser rendering, form submission, or payment execution. This provides a common set of rules for comparison, but leaves interactive and authenticated uses outside the study. The development record describes combining an AI-standards checklist with a general website checklist and later expanding its scope to AI search, agents, search engine optimization (SEO), and citations [34]. The 61-item version has a recorded commit date before the September measurements.

The exact count depends on how observations are grouped. Detecting a basic MCP response and finding its capabilities field count as two checks, while the two crawler-named requests produce one combined check. The six category totals below add up to 61. This is a defined, exploratory inventory, not a proven minimum or optimal number of readiness requirements. The supplementary archive gives the development history and the purpose, source basis, and limits of every check.

Table 2 gives the purpose of each category. Headers are information the server sends alongside the content.

Table 2: Purposes and sizes of the six checklist categories.

Category	**Purpose**	**Checks**
Discovery	Find resource lists, crawler rules, and guidance files	13
Access	Check response codes, available text, formats, and response headers	9
Interface and MCP	Look for possible service addresses, API descriptions, and access-permission information	10
Security	Look for security headers and contact details, without testing an agent's permission to act	11
SEO and structured data	Find titles, language tags, and descriptions of page content used by search engines and other software	15
Payments	Look for payment markers and terms relevant to commercial tasks	3

The set includes established formats, such as sitemaps and OpenAPI [5; 10], proposals such as llms.txt [6], and exploratory addresses or text rules. This explains why the observations may be relevant, but not whether each rule measures its intended feature accurately. Other checks within the same scope are possible. The study reports the complete recorded inventory, including items with few or no passes. Neither the development history nor the present data establish that these are the best 61 checks or that their total score predicts completed agent tasks.

3.2.1 Requests and matching rules

The following request details come from the archived implementation, whose check names and record format match the saved results. The records do not identify the exact executable used for each scan. Appendix B provides the source revision and complete matching rules.

For each domain, this implementation starts with 34 fetch operations and can try up to six additional resources. Following redirects can produce more network requests than this count. The requests cover the HTTPS homepage and addresses such as /robots.txt, /llms.txt, and /openapi.json, with a 4.5-second timeout for each fetch operation. One homepage request asks for Markdown or plain text.

Two others carry GPTBot and PerplexityBot names in the User-Agent header, which identifies the requesting software. The procedure does not log into sites or run page JavaScript. Several checks may use the same reply. A pass means that the reply or another input matches the rule for that check.

The text check uses the ordinary homepage reply. If that reply has no content, it uses the reply to the Markdown request. It removes scripts, styles, and HTML tags, then counts the remaining text. More than 300 characters produces a pass. Hidden text and error messages can contribute to that count.

The bot-request check records whether both, one, or neither of the named requests return HTTP 200 ("OK"). The names are labels in the test requests, not proof that either company's crawler made them. The MCP check sends an initialization request, the first step in starting an MCP session, to two possible service addresses. It looks for the text jsonrpc in a reply with a status code from 200 through 299. A further check tries to read the reply as JSON and looks for a capabilities field.

3.3 Saved results and repeat scans

Each saved domain record contains 61 check labels, an overall score, category scores, the scan time, and how long the scan took. The labels are pass, warning, fail, and not applicable. A warning can mean a partial match or a signal that was not found. A not-applicable label can mark an optional signal that was not found or a check that depends on another result. It does not necessarily mean that the check or its underlying request was skipped. Appendix C explains the scores and sampling adjustments.

The final dataset contains one record for each of the 50,000 sampled domains. Its retained timestamps span 18.77 hours on September 3-4, 2026 UTC. It includes newer results for all 21,547 domains whose earlier scores were 10 or below. Newer records replaced older ones even when scores fell. The final dataset therefore combines observations from different attempts under an outcome-dependent recheck policy.

3.4 Tests with prepared responses

After data collection, the study applied the archived rules to 13 prepared test cases, including empty replies, error pages, and ordinary web pages. Because the inputs were known, these cases showed when a rule could give a misleading result. These tests examine implementation behavior, not the unavailable historical replies from the sampled domains.

3.5 Comparisons

The comparisons below were chosen after the recorded results were available and are exploratory. Each comparison counts domain names. A closer analysis examines three checks, for llms.txt guidance files, Markdown responses, and the basic MCP marker. These represent guidance, an alternate content format, and a possible service interface. The analysis counts how many domains pass none, one, two, or all three checks, then compares the results across rank groups.

The access comparison counts domains that pass the text check and receive HTTP 200 for both named requests. The results come from separate requests. They show that the conditions occur on the same domain, not that one agent completed a task.

The CrUX comparison accounts for the fact that listed and unlisted domains may have different mixes of popular sites. It first compares the two sets within each rank group. It then combines those results as if both sets had the same mix, with 20% from the highest-ranked group, 40% from the middle group, and 40% from the lowest-ranked group. These are the shares in the sample. A second calculation uses 1%, 9%, and 90%, the shares in the full million-domain list. If the same group has the higher pass rate in both calculations, that result holds under either mix.

For repeat scans, the analysis matches each domain's older and newer records. It counts improvements, declines, and unchanged results. A change in the recorded result does not by itself show that the website changed.

4 Readiness Results

4.1 Infrastructure indicators

All 50,000 sampled domains have a final record with all 61 check labels. The average stored score is 25.7 out of 100. The median, or middle score, is 22. These scores summarize the original checklist and have not been calibrated to task success. Table 3 shows eleven checks from the six categories. Every percentage uses all 50,000 domains, including records labelled warning, fail, or not applicable. Appendix A gives the full counts.

Table 3: Selected check results across all 50,000 domains.

Check	Pass count	Pass share
robots.txt rule check	21,881	43.76%
llms.txt check	3,310	6.62%
Markdown-like response	1,474	2.95%
HTTP 200 for both test requests	18,912	37.82%
MCP response marker	910	1.82%
OpenAPI version marker	78	0.16%
security.txt contact marker	2,288	4.58%
JSON-LD script check	10,687	21.37%
Sitemap and robots.txt markers	13,686	27.37%
index.md body check	5,337	10.67%
Payment-related markers	163	0.33%

Both crawler-named requests were recorded as HTTP 200 on 18,912 domains (37.82%), one on 3,901 (7.80%), and neither on 27,187 (54.37%). The final category includes both non-200 responses and failed requests. The saved labels do not distinguish their causes or identify which request succeeded in the one-success group.

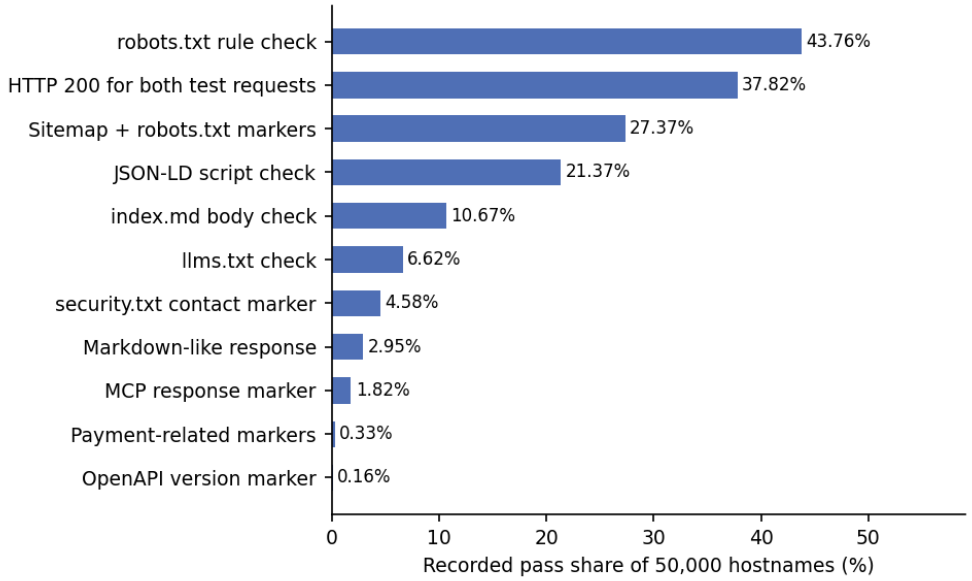


Figure 1: Percentage of all 50,000 sampled domains passing each of eleven checks.

4.1.1 How often the three selected checks pass together

At least one of the three selected checks passes on 3,748 domains (7.50%), while all three pass on 833. Most domains with any pass have only one of the three, but most MCP matches occur in the group passing all three. Table 4 counts each domain once.

Table 4: Number of selected checks passing per domain, with all 50,000 as the denominator.

Checks passing together	Domains	Share
0	46,252	92.50%
1	2,635	5.27%
2	280	0.56%
3	833	1.67%

Those 833 domains account for 91.54% of all MCP-marker matches. The Markdown check passes on 32.60% of domains passing llms.txt, compared with 0.85% of the remaining domains. Adding the three separate pass counts would count some domains more than once. After adjusting for the rank mix in the full million-domain list, the estimated share passing all three is 2.97%, and the share passing none is 91.91%.

4.2 Content access and request outcomes

The text-length check passed on 22,855 domains (45.71%) and failed on 27,145 (54.29%). Among domains passing it, both crawler-named requests returned HTTP 200 on 65.98%, one on 11.28%, and neither on 22.74%. The average checklist score in this group was 41.1.

Both access conditions occur on 15,080 domains, or 30.16% of the full sample. Requiring both conditions gives fewer matches than checking either alone. The returned content still needs inspection because the text rule can accept an error page.

Domains failing the text check still produced positive results on other checks. For example, 452 passed the llms.txt check and 2,231 passed the sitemap-and-robots check. Table 5 reports counts and percentages within each group; a domain can appear in several rows.

Table 5: Other check results conditional on the text-length label.

Check	Text pass (n=22,855)	Text fail (n=27,145)
HTTP 200 for both test requests	15,080 (65.98%)	3,832 (14.12%)
llms.txt check	2,858 (12.50%)	452 (1.67%)
Markdown-like response	1,452 (6.35%)	22 (0.08%)
Sitemap and robots.txt markers	11,455 (50.12%)	2,231 (8.22%)
JSON-LD script check	10,446 (45.71%)	241 (0.89%)
MCP response marker	890 (3.89%)	20 (0.07%)

The gap for the two-request check is 51.86 percentage points. Domains passing the text rule pass other checks much more often, so results from that group alone cannot describe the full sample. Some links between the checks also follow from how they work. The text and JSON-LD checks read the same selected page content, and the text check can use the Markdown reply if the ordinary reply is empty.

Keeping only domains that pass the text check would discard 13.66% of the llms.txt passes, 16.30% of the sitemap-and-robots passes, and 20.26% of the both-request HTTP-200 passes. Even when the text check fails, other addresses or request types can still produce positive labels. All domains remain in the main analysis.

4.3 Readiness profiles by rank

Final average scores differ by less than one point across the three rank groups. Before the low-score rechecks, the averages were 20.7, 22.4, and 23.9. The rechecks covered 52.44%, 43.12%, and 38.40% of the groups, respectively.

Table 6: Stored scores and request outcomes within each rank group.

Rank group	Domains	Average score	Median	Neither test returned HTTP 200
1-10,000	10,000	26.1	23	56.25%
10,001-100,000	20,000	25.2	20	55.58%
100,001-1,000,000	20,000	25.9	22	52.23%

Similar averages hide differences in the individual checks. Markdown and MCP-marker rates rise from the highest- to the lowest-ranked group, while the llms.txt rate is lower in the middle group. The basic MCP counts are six, 219, and 685. Two groups can have similar overall scores but different kinds of positive results.

Table 7: Selected pass rates within each rank group.

Rank group	llms.txt check	Markdown-like response	MCP response marker
1-10,000	6.38%	1.66%	0.06%
10,001-100,000	5.86%	2.48%	1.10%
100,001-1,000,000	7.51%	4.07%	3.43%

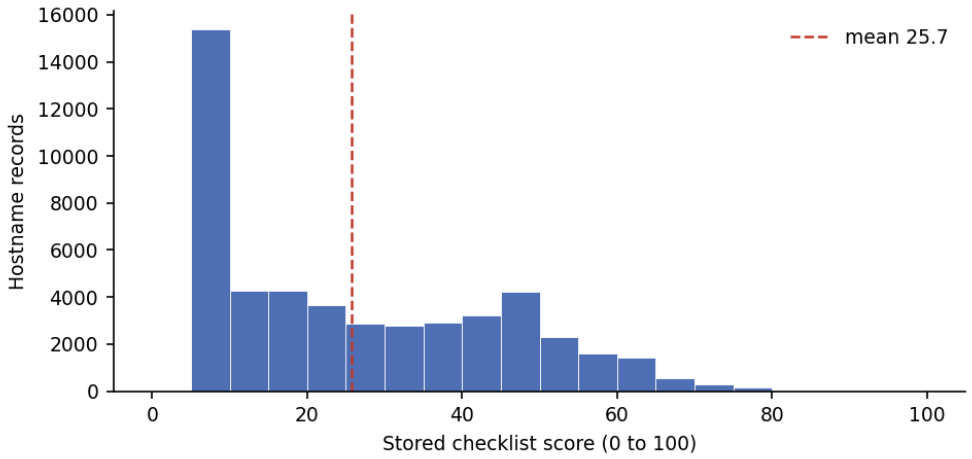


Figure 2: Number of domains in each score range, with the average marked.

4.3.1 Adjusting for the mix of ranks

Highly ranked domains make up a larger share of the sample than of the full million-domain list. The adjusted results give each rank group its share in that list. They estimate the results of this collection procedure for the Tranco frame, assuming comparable measurement conditions. Table 8 includes all nine indicators in the readiness profile, the two headline combinations, and the stored score and neither-request outcome. Earlier and final estimates show how selective rechecking affects the profile.

The brackets show approximate 95% ranges for uncertainty from sampling only some domains, treating the recorded values as fixed. They omit request variability, mistaken matches, and uncertainty about selective rechecking. Each range concerns one measure; it is not a test of differences between checks or groups. Appendix C explains the adjustment.

Table 8: Sample values and rank-adjusted estimates for both record versions. Brackets give individual sampling-only 95% ranges.

Measure	Final sample	Earlier, adjusted	Final, adjusted
Checklist score	25.69	23.77 [23.53, 24.00]	25.87 [25.64, 26.10]
Neither test returned HTTP 200	54.37%	57.57% [56.96%, 58.18%]	52.57% [51.95%, 53.19%]
Text-length check	45.71%	41.41% [40.79%, 42.02%]	46.58% [45.96%, 47.20%]
HTTP 200 for both test requests	37.82%	33.88% [33.29%, 34.46%]	38.11% [37.51%, 38.71%]
Text check and both HTTP-200 replies	30.16%	26.60% [26.05%, 27.15%]	29.91% [29.35%, 30.48%]
Sitemap and robots.txt markers	27.37%	24.57% [24.03%, 25.10%]	27.54% [26.98%, 28.09%]
JSON-LD script check	21.37%	18.44% [17.96%, 18.92%]	20.67% [20.16%, 21.17%]
llms.txt check	6.62%	6.60% [6.29%, 6.91%]	7.35% [7.02%, 7.67%]
Markdown-like response	2.95%	3.54% [3.31%, 3.77%]	3.90% [3.65%, 4.14%]
MCP response marker	1.82%	2.91% [2.69%, 3.12%]	3.18% [2.96%, 3.41%]
OpenAPI version marker	0.16%	0.09% [0.05%, 0.13%]	0.10% [0.06%, 0.13%]
At least one of three selected checks	7.50%	7.26% [6.94%, 7.59%]	8.09% [7.75%, 8.43%]
Payment-related markers	0.33%	0.29% [0.22%, 0.36%]	0.33% [0.26%, 0.40%]

The adjustment raises the final MCP marker rate from 1.82% to 3.18%. More matches occur in the lowest-ranked group, and that group makes up most of the full list. The average score changes much less. The mix of ranks can therefore matter for an individual check even when the overall score barely changes.

The joint text-and-both-request estimate is 29.91%, compared with 30.16% in the sample. The share passing at least one selected check rises from 7.50% to 8.09%, while the OpenAPI-marker estimate falls from 0.16% to 0.10%. Weighting does not move every estimate in the same direction. The text, sitemap, and JSON-LD checks still have higher match rates than the selected guidance and interface checks under either rank mix.

4.4 Comparison with the CrUX list

The 23,592 domains found in the CrUX list pass each of the five checks below more often than the other domains. The first two result columns show the observed percentages, with pass counts in parentheses. The last column compares the groups after giving them the same rank mix, as explained in Section 3.5. It is based on the adjusted rates, not a subtraction of the first two columns. Gaps are in percentage points (pp).

Table 9: CrUX comparisons before and after giving both groups the sample's rank mix.

Check	In CrUX (n=23,592)	Outside CrUX (n=26,408)	Gap with same rank mix (pp)
llms.txt check	8.86% (2,091)	4.62% (1,219)	+4.66
MCP response marker	2.83% (668)	0.92% (242)	+2.54
HTTP 200 for both test requests	45.78% (10,800)	30.72% (8,112)	+15.64
Markdown-like response	4.02% (948)	1.99% (526)	+2.50
robots.txt rule check	54.57% (12,875)	34.10% (9,006)	+21.22

The CrUX-listed domains have higher pass rates within every rank group. They also have higher rates with either of the two rank mixes. For the bot-request check, the gap is 15.06 percentage points before adjustment and 15.64 after it. The broad rank mix alone therefore does not explain the difference between listed and unlisted domains.

This does not show that more traffic causes higher pass rates. CrUX also has rules about whether pages can be found publicly. Site type and differences within the broad rank groups may also matter, but this study does not measure them.

4.5 Where the checks can mislead

Some rules counted unrelated content as a match. Returning the same ordinary HTML page from every tested address produced 23 passes and a score of 47. A long HTTP 403 ("Forbidden") page passed the text check. A short page that needed no JavaScript failed it.

Table 10: Selected prepared-response tests and their recorded outcomes.

Test response	**Result**
Every request raises an error	All 61 labels are produced; score 6
Long HTTP 403 error page	Text check passes
Short static HTTP 200 page	Text check fails
Same generic HTML at every path	23 passes; score 47
HTTP 200 error reply containing jsonrpc	Basic MCP check passes
Capabilities reply wrapped as a server-sent event	Basic MCP check passes; capabilities check warns
Wildcard robots rule disallowing all paths	robots.txt rule check passes

Failed requests and empty HTTP 404 ("not found") replies produced the same 61 labels in the tests. The same pattern appears in 14,776 real records (29.55%). The saved labels cannot tell which of those conditions, or another condition with the same results, occurred.

Of the 910 basic MCP matches, 3 also passed the JSON-and-capabilities check, while 907 received a warning. The two checks may read different replies. The capabilities check tries to read the entire reply as one JSON document. It misses the capabilities field when the JSON is wrapped in Server-Sent Events (SSE) messages [30]. A prepared test reproduced this failure while the basic MCP marker check still passed. A warning therefore cannot automatically be read as an invalid service.

4.6 How repeat scans affect the comparison

The repeat scans raised the average score from 22.68 to 25.69, a change of 3.00 points across the full sample. The change uses the unrounded means. Among the 21,547 rechecked domains, scores increased on 5,965, decreased on 553, and remained unchanged on 15,029. Most rechecked scores therefore did not improve, even though the average rose.

A new pass means that a check did not pass before but passed on the repeat scan. A lost pass means the reverse. The last column shows the change in the pass rate across all 50,000 domains, in percentage points. The text and bot-request checks changed more than the three selected resource checks.

Table 11: Paired pass gains and losses under the selective recheck policy.

Check	**Pass gains**	**Pass losses**	**Net change (pp)**
Text-length check	3,629	0	+7.26
HTTP 200 for both test requests	3,018	16	+6.00
llms.txt check	467	5	+0.92
Markdown-like response	171	0	+0.34
MCP response marker	84	0	+0.17

The text-check count rose from 19,226 (38.45%) to 22,855 (45.71%). Only one rechecked domain had originally passed this check. There was therefore almost no chance to observe a pass becoming a failure in that group. Zero lost passes does not show that the result would stay the same over time. Repeating scans across the full score range could give a fairer view of gains and losses.

The highest-ranked group gains 5.37 score points, compared with 1.99 in the lowest-ranked group. This reverses which group has the higher average. The repeat scans covered a larger share of the highest-ranked group, and the average gain among its rechecked domains was also larger, 10.24 versus 5.19 points. The final comparison therefore depends partly on which domains were tried again. It does not show that the groups have equal capabilities or that websites became more capable over time.

5 What the Results Mean for Agent Readiness

The results show different match rates across the selected checks. The text, sitemap, and JSON-LD rules pass more often than the selected guidance and interface checks. This compares results from rules with different strengths and weaknesses; it does not establish that the same proportions of websites offer useful content or working interfaces. Each percentage below uses all 50,000 domains.

Table 12: Recorded sample evidence by activity and its interpretation.

Readiness area	Recorded evidence	Assessment
Content access	Text check 45.71%; text check and both "OK" replies 30.16%	Matches to two recorded conditions; content usefulness was not tested
Resource discovery	Sitemap-and-robots check 27.37%; JSON-LD 21.37%; llms.txt 6.62%	Common web formats pass more often than the dedicated guidance-file check
Service interaction	MCP marker 1.82%; OpenAPI marker 0.16%; payment markers 0.33%	Few matches at the tested addresses; successful actions were not tested

5.1 Reading and discovering information

The differing matches motivate testing more than one route to a resource. These routes can include a page, a sitemap, structured metadata, or a separate guidance file. The comparison does not show which route yields the most accurate answer.

Most MCP matches occur on domains that also pass the llms.txt and Markdown checks. Sites may publish those resources together, or their servers may return similar content from several addresses. The saved results cannot tell how much each explanation contributes. The overlap shows where the checks agree, but not whether the matched content is useful for a task.

A domain with little homepage text can still pass a check at another address. A long error page can also pass the text rule. For that reason, the amount of homepage text alone is a poor filter for all the other observations. Each requested resource needs its own check.

5.2 Moving from information to action

The sparse MCP and OpenAPI matches describe the reach of those particular address and matching rules. A service may still exist elsewhere on the website or return a format that the rule misses. Finding it could require reading documentation, trying another address, or using a full browser, which these checks did not do.

The study did not test whether an agent could use an advertised operation. The event-stream example in Section 4.5 also shows an allowed reply format that the capabilities rule misses. Positive and negative labels therefore concern matching conditions rather than functionality. Security headers, access-permission descriptions, and payment-related words cannot fill the gap in task outcomes.

5.3 Priorities and task-based follow-up

Website owners can evaluate retrievable pages, lists of important URLs, and clear API documentation as candidates for supporting agents. Services can describe their operations and required permissions. Publishing a guidance file alone does not show that an agent uses it, as the Ahrefs request counts illustrate [26].

Agent developers can test whether clients benefit from combining common web formats with agent-specific files. Content validation is a separate step. Thunderbit rejected many HTTP-200 replies because they did not contain the intended llms.txt file [22]. Distinguishing failed requests, unrelated pages, and unsupported operations provides information for choosing whether to retry, use another address, or stop.

The association with rank depends on the check. Similar average scores hide different profiles, and rechecking low scores changes which rank group has the higher average. CrUX membership provides another comparison, but does not show what a particular service permits. The relevant question is which features the intended task needs and whether the target site provides them.

A task-based follow-up could compare direct-request and browser agents on the same information-finding and authorized service-use tasks. It needs separately verified task outcomes, domains with and without check matches, and retained request replies and errors. Keeping check development separate from the task evaluation permits testing whether the signals predict success. This is a proposed evaluation, not an outcome of the present study.

6 Limitations

The saved records contain check labels, but not the full replies, individual request codes, detailed errors, redirect histories, or points awarded by each rule. This prevents later analysis from sorting failures by cause, testing a different minimum text length, or independently rebuilding every score. The prepared tests show how the rules can mislead, but not how often that happened among the 50,000 domains.

The checklist includes ordinary website features and optional formats as well as possible service interfaces. These checks need not matter equally for an agent's task. The study has not shown that the set is complete or that passing more checks makes an agent more likely to succeed.

The findings also depend on the requested addresses, request labels, timeout, and rules for trying another response. The crawler-named requests were not verified visits by those crawlers. The MCP request used fixed settings. Alongside the missing executable identifier noted in Section 3.2.1, the original scan network setup was not retained. The recheck policy prevents interpreting the final records as a uniform single-attempt snapshot or a general test of temporal reliability.

Some Tranco domains serve technical systems rather than public pages. The sample does not label site type or industry. The text and keyword rules were not tested for equal accuracy across languages and page types. Logins, private APIs, other internet protocols, and tasks performed through a full browser are outside the study. The percentages describe recorded check results under these request settings; they are not verified adoption rates for the whole web.

7 Conclusion

The study finds uneven matches across the selected public-web checks in the September 2026 sample. The text check passes on 45.71% of domains. Across all domains, 30.16% pass that check and receive HTTP 200 for both crawler-named requests. At least one of the three checks for llms.txt, Markdown, and a basic MCP marker passes on 7.50% of domains. These are recorded check results, not a validated readiness grade for the web.

The scan finds basic MCP and OpenAPI markers on 1.82% and 0.16%, respectively, at the tested addresses. The study did not execute complete tasks. Its practical contribution is a reproducible profile of access, discovery, and interface checks, together with evidence that rank weighting and selective rechecks change that profile. Verified content and task outcomes are needed to establish actual use; the internet's overall agent readiness remains unmeasured.

Data and Code

The accompanying archive contains the domain records, sampling inputs, checking-program source, analysis scripts, and software tests. It also includes the checklist's development record and a check-by-check explanation of purpose and limits, with software versions and file checksums so readers can verify their copies.

Appendix A. Complete Check Results

Each row sums to 50,000. Appendix B explains the labels. NA means not applicable and includes optional checks and checks that depend on another result.

Check	Pass	Warning	Fail	NA
access-boilerplate-ratio	7,737	42,263	0	0
access-compression	28,210	21,790	0	0
access-js-hydration	22,855	0	27,145	0
agent-card	3,791	46,209	0	0
agent-payments	163	0	0	49,837
agents-json	3,828	46,172	0	0
agents-txt	4,136	45,864	0	0
ai-catalog	3,890	46,110	0	0
api-catalog-rfc9727	3,883	46,117	0	0
api-cors-ai	3,215	46,785	0	0
api-dry-run	13	5,559	0	44,428
ard-catalog	20	3,566	46,414	0
auth-guide	4,372	45,628	0	0
bot-ua-access	18,912	3,901	27,187	0
canonical	15,771	0	34,229	0
csp	9,661	40,339	0	0
favicon-branding	26,086	23,914	0	0
frame-protection	16,580	33,420	0	0
hsts	14,243	35,757	0	0
html-lang	25,119	24,881	0	0
https-tls	50,000	0	0	0
json-ld	10,687	0	39,313	0
link-headers	2,173	47,827	0	0
llms-full	5,771	44,229	0	0
llms-txt	3,310	5,418	41,272	0
markdown-negotiation	1,474	0	48,526	0
markdown-twins	5,337	44,663	0	0
mcp-schema-handshake	3	907	0	49,090
mcp-jsonrpc-screen	910	49,090	0	0
meta-description	13,634	5,657	30,709	0
oauth-agent-discovery	5,709	44,291	0	0
og-tags	14,486	0	35,514	0
openai-plugin	4,817	45,183	0	0
openapi-examples	50	5,522	0	44,428
openapi-spec	78	49,922	0	0
payments-terms	4,738	45,262	0	0
payments-webln	30	0	0	49,970
permissions-policy	4,962	45,038	0	0
rate-limit-headers	361	49,639	0	0
referrer-policy	10,313	39,687	0	0
rfc-7807-errors	6	0	0	49,994
robots-ai-policy	21,881	3,638	24,481	0
robots-meta-ai	46,390	0	3,610	0
robots-selective-ai-policy	4,420	21,099	24,481	0
rss-feed	3,365	46,635	0	0
security-ai-disclosure	23,382	26,618	0	0
security-c2pa	1	49,999	0	0
security-signatures	4,096	45,904	0	0

Check	Pass	Warning	Fail	NA
security-txt	2,288	47,712	0	0
seo-answer-first	15,310	34,690	0	0
seo-author-eeat	6,910	43,090	0	0
seo-freshness	3,632	46,368	0	0
seo-multimodal	47,814	2,186	0	0
seo-rich-schemas	6,230	43,770	0	0
seo-schema-graph	8,474	41,526	0	0
sitemap-json	2,903	47,097	0	0
sitemap-xml	13,686	3,581	32,733	0
tdmrep	3,514	46,486	0	0
title-tag	25,103	4,192	20,705	0
twitter-cards	11,871	38,129	0	0
xcto	14,596	35,404	0	0

APPENDIX B. CHECK DEFINITIONS

These tables explain exactly what produces a pass. GET asks for content and POST sends data. The response body is the content returned, while headers give information alongside it. HTTP 2xx means any status from 200 through 299. A nonempty body contains at least one character. JSON is a structured text format; parsing means reading its structure. A regex is a text-matching pattern.

The tables keep exact field names and paths so the rules can be reproduced. Each path starts at the domain's HTTPS address unless a reply gives another URL to try. The check named mcp-jsonrpc-screen is stored as mcp-server-live in the data. The interface category is stored as usability.

B.1 DISCOVERY

The discovery checks look for crawler rules [4], sitemaps [5], guidance for language models [6], and API lists [7]. ARD is the program's chosen address for a possible Agent Readiness Directory. The llms.txt check requires more than the proposal's minimum project heading and tries only the listed addresses.

Check	Request	Match rule
robots-ai-policy	GET /robots.txt	Nonempty HTTP-2xx body; no regex match for root-wide Disallow after GPTBot, or no such match after ClaudeBot
robots-selective-ai-policy	GET /robots.txt	Nonempty HTTP-2xx body contains a token from each of two scanner-defined bot-name lists
llms-txt	GET /llms.txt; try /.well-known/llms.txt after failure, non-2xx, or empty body	HTTP-2xx body over 50 chars with an H1-style line and Markdown link syntax
llms-full	GET /llms-full.txt and /llms-small.txt	HTTP-2xx full-file body over 100 chars, or any nonempty HTTP-2xx small-file body
ard-catalog	GET /.well-known/ard.json	Nonempty HTTP-2xx body parses as JSON
ai-catalog	GET /.well-known/ai-catalog.json	Any HTTP-2xx response
api-catalog-rfc9727	GET /.well-known/api-catalog	Nonempty HTTP-2xx body
sitemap-xml	GET /sitemap.xml and a robots-declared URL; /sitemap_index.xml after root-sitemap failure or empty body	Selected HTTP-2xx body contains <urlset or <sitemapindex; robots text also has a Sitemap HTTP-URL directive
sitemap-json	GET /.well-known/sitemap.json	Nonempty HTTP-2xx body contains {
agents-txt	GET /agents.txt	HTTP-2xx body over 20 chars
agents-json	GET /.well-known/agents.json	Any HTTP-2xx response
agent-card	GET /.well-known/agent-card.json	Any HTTP-2xx response [9]
openai-plugin	GET /.well-known/ai-plugin.json and /plugin.json	Any HTTP-2xx response at either path

B.2 Access

These checks inspect response codes, returned text, and headers. They also ask the server for particular content formats, a process called content negotiation [13], and examine the Link header [14].

Check	Request	Match rule
markdown-negotiation	GET / with Accept: text/markdown, text/plain;q=0.9	HTTP-2xx; Content-Type contains text/markdown or the body matches the H1/front-matter regex
markdown-twins	GET /index.md	HTTP-2xx body over 50 chars
bot-ua-access	GET / with GPTBot and PerplexityBot user-agent strings	Both requests classified as HTTP 200
robots-meta-ai	Selected meta robots, meta googlebot, and X-Robots-Tag	No noindex or nosnippet substring; extra points for specified snippet or image-preview markers
link-headers	Selected Link header	Exact double-quoted rel value describedby, service-desc, or alternate
rate-limit-headers	Selected rate headers	Nonempty ratelimit-limit, x-ratelimit-limit, ratelimit-remaining, or Retry-After value
access-js-hydration	Selected root body, or Markdown-request body as fallback	More than 300 characters after script, style, tag, and whitespace processing
access-compression	Selected content-encoding header	Lowercased value contains br, zstd, or gzip
access-boilerplate-ratio	Extracted text length divided by selected body length	Rounded percentage at or above 10%

B.3 Interfaces

These checks look for a reply to an MCP connection request [8], an OpenAPI description of an API [10], OAuth information about access permissions [11; 12], and a TDMRep file about text-and-data-mining preferences [20]. Some rules below use JavaScript's idea of a value being true, called "truthy". For example, a nonempty string or an object counts as true, while a missing value, an empty string, or false does not.

Check	Request	Match rule
mcp-jsonrpc-screen	POST initialize to /.well-known/mcp and /api/mcp	Either HTTP-2xx body contains jsonrpc
mcp-schema-handshake	After a basic MCP pass, parse the first nonempty body in well-known then API-path order	JSON parses; Boolean(result.capabilities) is true
openapi-spec	GET /openapi.json and /.well-known/openapi.json; try /openapi.yaml if neither fetch is HTTP-2xx; select first qualifying body over 50 chars	Parsed JSON has a truthy openapi or swagger value; on parse failure, body contains openapi: or swagger:
openapi-examples	Selected OpenAPI-candidate text	Case-insensitive match for "example": or "examples":
auth-guide	GET /auth.md	HTTP-2xx body over 30 chars
oauth-agent-discovery	GET /.well-known/oauth-authorization-server and /.well-known/oauth-protected-resource	Any HTTP-2xx response at either path
tdmrep	GET /.well-known/tdmrep.json	Nonempty HTTP-2xx body
rfc-7807-errors	GET /api	Content-Type contains application/problem+json
api-dry-run	Selected OpenAPI-candidate text	Case-insensitive quoted dry_run, dryRun, validate_only, idempotency_key, or Idempotency-Key token
api-cors-ai	First truthy Access-Control-Allow-Origin value in root, /api, then /openapi.json order	Selected value remains nonempty after trimming

B.4 SECURITY

This group looks for security-related headers such as HSTS [15] and Content Security Policy [16], and security contact information in security.txt [17]. The HTTPS check only reads the start of the input URL. It therefore passes for every domain in this sample without testing the connection itself.

Check	Request	Match rule
https-tls	Normalized input scheme	Input begins with HTTPS
hsts	Selected Strict-Transport-Security	Contains max-age
csp	Selected Content-Security-Policy	Value longer than 10 chars
xcto	Selected X-Content-Type-Options	Lowercased value contains nosniff
frame-protection	Selected headers	Nonempty X-Frame-Options, or CSP contains frame-ancestors
referrer-policy	Selected Referrer-Policy	Nonempty value
permissions-policy	Selected Permissions-Policy	Nonempty value
security-txt	GET /.well-known/security.txt; try /security.txt after failure, non-2xx, or empty body	HTTP-2xx body contains Contact:
security-signatures	GET /.well-known/http-message-signatures-directory	Any HTTP-2xx response
security-c2pa	Selected root-body text	Contains application/c2pa or c2pa.org
security-ai-disclosure	Bodies from /.well-known/security.txt and /terms-of-use.md	Regardless of HTTP status, case-insensitive AI/LLM/Agent/Prompt-Injection or AI/Safety/Model substrings, respectively

B.5 METADATA

Metadata describes a page for other software. These checks include content labelled as JSON-LD [18] and terms associated with Schema.org [19].

Check	Request	Match rule
title-tag	Regex-selected title-tag text	Trimmed length from 10 through 75 chars
meta-description	Selected meta-description text	Length from 50 through 170 chars
canonical	Selected canonical-link value	Starts with http://, https://, or /
html-lang	Selected html lang attribute	Trimmed value has at least two chars
favicon-branding	GET /favicon.ico and selected root body	HTTP 200 from icon path, or a matching icon-link marker
og-tags	Selected Open Graph values	Nonempty og:title and either og:description or og:image
twitter-cards	Selected Twitter values	Nonempty twitter:card or twitter:title
json-ld	Script-content regex with an application/ld+json marker	At least one selected block parses as JSON
seo-schema-graph	Selected root-body text	One of "@graph", "isPartOf", "mainEntity", or "author"
seo-rich-schemas	Selected root-body text	Case-insensitive quoted FAQPage, HowTo, Article, BlogPosting, SoftwareApplication, WebApplication, Course, DefinedTermSet, or BreadcrumbList
seo-author-eeat	Selected root-body text	Contains "sameAs", wikidata.org, orcid.org, scholar.google.com, or both "Person" and "name"
seo-answer-first	Selected root-body text	H1 and H2 tag regexes both match
seo-multimodal	Selected root-body text	No lowercase img opening marker, or at least one image matches nonempty alt text
seo-freshness	Selected root-body text	Case-insensitive last_updated, lastmod, published_time, or nonempty quoted dateModified/datePublished value

Check	Request	Match rule
rss-feed	GET /feed.xml and /feed.json plus a root-body link URL when found; prefer qualifying XML, then any HTTP-2xx JSON response, then the declared response	Selected body contains <rss, <feed, "version", or the exact quoted Atom namespace marker

B.6 Payments

Check	Request	Match rule
agent-payments	Selected headers and root-body text	L402/LSAT authentication marker, nonempty x-payment header, or x402/lightning: text
payments-webln	Selected root-body text	Contains webln or lightning:
payments-terms	GET /terms-of-use.md and selected root body	Any HTTP-2xx from terms path, or terms-of-use.md or ai-terms text

B.7 Response selection

Text checks use the ordinary homepage reply if its body contains any characters. Otherwise, they use the Markdown reply, or empty text if neither has content. Several header checks use the first available header set in this order, ordinary homepage, Markdown request, then GPTBot request. The program makes these choices without requiring a successful response code. Text and header checks can therefore read different replies.

The text rule removes scripts, styles, and tags using text patterns, then reduces repeated spaces and line breaks. It uses JavaScript's string length, measured in UTF-16 code units, so some characters count as two. Hidden text also counts, and a written HTML code such as & stays encoded rather than becoming one displayed character. Failed requests give the checks no response object, and failures while reading a reply give them empty text.

B.8 Request settings

The rules follow scanner revision ea0e797. That version predates the saved results and uses the same check names and record format.

Most requests use VedaAgenticScanner/1.0 (+https://veda.ng/scan; agent-readiness-audit). The Markdown root request adds Accept: text/markdown, text/plain;q=0.9. The two bot root requests use Mozilla/5.0 (compatible; GPTBot/1.0; +https://openai.com/gptbot) and Mozilla/5.0 (compatible; PerplexityBot/1.0; +https://perplexity.ai/perplexitybot).

MCP requests use POST and Content-Type: application/json. Their JSON-RPC 2.0 initialize payloads specify protocol version 2024-11-05, empty capabilities, and client name veda-scanner version 1.0. Request IDs are probe-1 and probe-2. The implementation stops after the response and lacks the Accept-header and event-stream handling required by the later Streamable HTTP transport [30].

The program follows redirects. It does not separately try adding www or changing HTTPS to HTTP. A sitemap or feed link can lead to another host, but the result still counts under the starting domain. Requests do not log in. The program fetches robots.txt alongside the other resources instead of checking its rules before making requests. It does not submit forms, make payments, or keep cookies between requests. The watchdog allows three domains to be processed at once in each process and can run several processes.

Appendix C. Scores and Sampling Weights

C.1 Checklist score

The program compares the points earned with the maximum points from checks it marks neither optional nor not applicable. It expresses the result as a percentage, rounds to a whole number, and caps it at 100. Optional checks can add points without increasing the maximum used in this calculation. If that maximum is zero, the program uses one instead.

Two results with the same label can earn different points, so this paper uses the scores saved by the program. The input-HTTPS rule adds three points per domain. Category scores use the same method. Their averages are 41.0 for security, 28.9 for metadata, 23.9 for access, 20.7 for discovery, 9.8 for payments, and 7.4 for interfaces.

Optional points can increase the score without increasing its denominator, while applicability changes which checks enter that denominator. Equal scores therefore need not represent equal sets of passed checks or equal capabilities. The score comparisons describe the retained scoring system; the individual labels carry the more specific observations.

C.2 Why the results are weighted

The sample gives highly ranked domains a larger share than they have in the full Tranco list [1]. Weighting means giving each rank group its share of that full list when combining the results. The top 10,000 contribute 1%, the next 90,000 contribute 9%, and the remaining 900,000 contribute 90%. These figures estimate outcomes under the same collection procedure for the full list, assuming comparable measurement conditions. They do not estimate all websites or remove the effect of selective rechecking. The separate CrUX comparison in Section 3.5 gives listed and unlisted domains the same chosen mix of ranks.

The bracketed 95% ranges use a normal approximation with a correction for the share of each rank group sampled. They describe uncertainty from sampling only some domains in the lower two groups. All top-10,000 domains were included, so there is no sampling uncertainty for that group. These are separate ranges for each measure, conditional on the saved labels. They do not include request variability, mistaken matches, or uncertainty about repeat scans, and do not provide a combined test across the checklist. The analysis code gives the detailed calculations.

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